Introduction to Data Analytics
MS - 4610

Session 7 – Distribution Fitting

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Role of applied statistics and probability in prediction

- An example: We are an e-commerce portal that needs to decide how much of a product we should stock. We know the latent demand for a product by the number of searches we receive each day. What other information do you need to figure out the answer?
  - Cost price. Is that fixed?
  - Transportation cost. Is it fixed? Price of fuel? Price of labour? Number of available employees? Likelihood that a guy will quit? Not show up?
  - Holding cost/cost of carry over
  - Model these aspects as random variables (from data), and then simulate different scenarios.
    - In some cases make decisions based off of it.

Fitting Distributions: Motivation

- Number of hear attacks per day:
  - The truth: $N(20,5^2)$
  - A dataset:

```
10 15 15 17 17 18 18 18 18 18 19 21 21 22 22 24 25 27 27 30
```

- What if the hospital wants to plan a capacity for 90% of the days
  - What is the correct solution?

```
qnorm(.9,mean = 20, sd = 5)
```

or norm.inv(0.9, 20,5) = 26.40776

  - What is the empirical solution?
  - By distribution modelling (using MLE) it created a $N(20.14,4.75^2)$
  - The distribution modelling solution: 26.22

Fitting Distributions

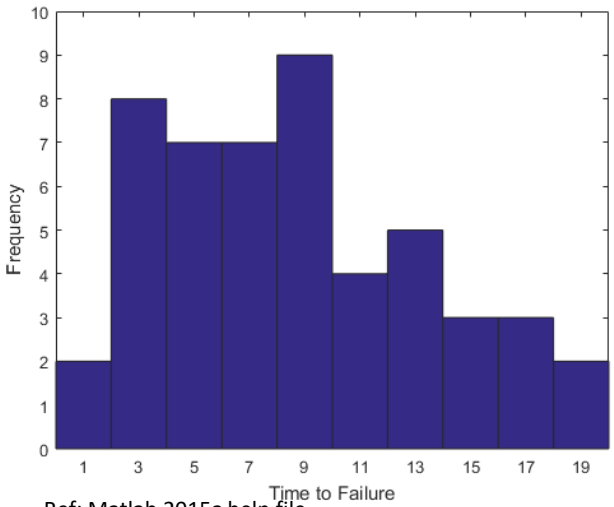
- Objective
- Distribution fitting versus curve fitting
- Steps
  - Model/function choice
    - Graphical
    - Analytical
  - Estimate Parameters
    - Analogic, method of moments
    - Maximum likelihood
    - Regression on Probability plots
  - Measures of Goodness-of-fit (statistical tests)

Model/function choice

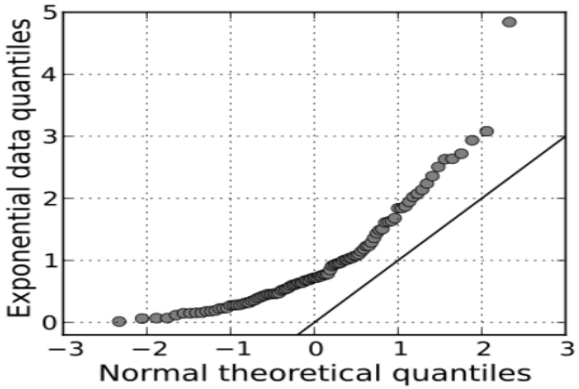
- Business insights
- Graphical
  - Histogram as a first step For visualization
  - Many formal methods might need bin sizes
  - CDFs
  - Normal Q-Q plot
- Analytical Solutions
  - Pearson's K criteria:

$$K = \frac{\gamma_1^2(\gamma_2+6)^2}{4(4\gamma_2 - 3\gamma_1^2 + 12)(2\gamma_2 - 3\gamma_1^2)}$$

$$\gamma_1 = \frac{\sum_{i=1}^{n}(x_i-\mu)^3}{n\sigma^3} \qquad \gamma_2 = \frac{\sum_{i=1}^{n}(x_i-\mu)^4}{n\sigma^4} - 3$$



Ref: Matlab 2015a help file



Ref: Q-Q plot in Wikipedia

Fitting Distributions

- Parametric approach: You know the distribution now you need to know the parameters
  - Analogic
  - Method of Moments.
    - Estimate the moments E(X), E(X^2), E(X^3)…, from the sample
    - Relate this to the parameters.
    - Solve for the parameters
    - e.g., Bernoulli or Binomial has parameter p, Normal has parameters μ and σ, Uniform has parameters a and b.
  - Maximum Likelihood estimation
    - Maximizes the independent likelihood of observing these values

$$\arg max\, \hat{l}(\theta; x_1, x_2 .. x_n) = \prod_{i=1}^{n} f(x_i|\theta)$$

  - Probability plots with regression Analysis (Principle only)

Statistical Tests once you know the parameters

- Chi-Square goodness-of-fit, Kolomgorov-Smirnov, Anderson-Darling Lillie, etc.
- Chi-Square GOF in detail (Also called pearson's Chi-Square test)
  - Hypothesis: Null: The data belongs to the specified distribution, alt: …
  - Test statistic is chi-squared with n degrees of freedom $\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$
  - How do we get O_i
  - How do we get $E_i = \left(F\left(Y_{ub(i)}\right) - F\left(Y_{lb(i)}\right)\right) * N$

Simulation

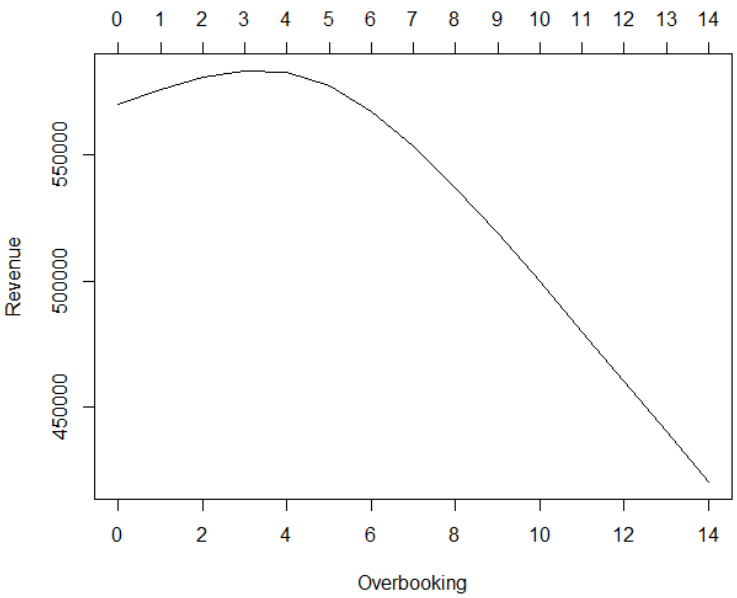
- Replicate the uncertainty in distributions to answer business questions. Uncertainty in Supply, demand, weather, flight timings, defaulting on loans, stock prices, competitors prices, etc.
- Why not use the distributions directly? What about complex dependencies?
- Toy problem: Bernoulli to Binomial
- Sample problem: A plane has 100 economy class tickets. Each ticket costs ₹ 6000. Not everyone who booked a ticket shows up. The distribution of people not showing up is Poisson Distributed with mean = 5. If they over-book they need to bump people up to first class/put them in a hotel, which costs the airline ₹20,000. How many tickets should you sell?

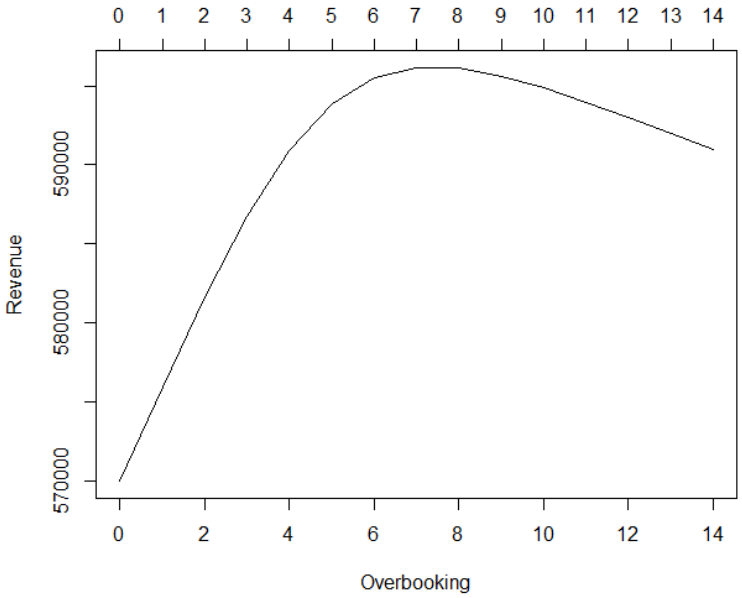
Steps

- What is the sequence of steps?
- What is the objective function?
- R-code:

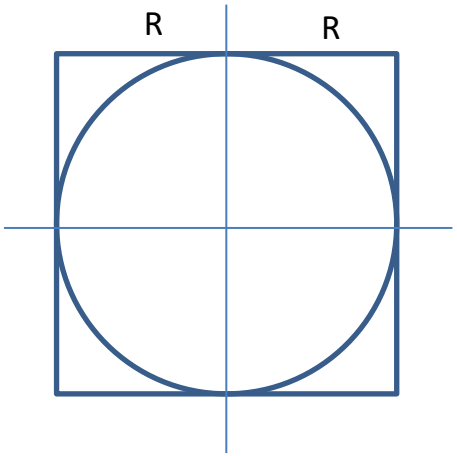
```
rm(list = ls())
noshows = rpois(10000,5)
revenue<-vector(length=15)
for (i in 1:15){
  showedup = 100+i-1-noshows
 # what we want: revenue_casebycase = 6000*(100+i-1)- 20000*max(0,showedup-100)
  revenue_casebycase = 6000*unlist(lapply(showedup,function(x) min(100,x))) -
20000*unlist(lapply(showedup,function(x) max(x-100,0)))
  revenue[i] = mean(revenue_casebycase)
}
```

Results





Contrast between 20k penalty and 1000

Simulation

- Can solve problems even with no uncertainty
- Derive π